Practice Set B

# Class 12 Mathematics, Chapter 2: Inverse Trigonometric Functions

Maximum marks: **43** **20** questions

Name _____ Date _____ Score _____ / 43

1. All questions are compulsory. Marks are shown against each question.
2. Show every step of working: steps carry marks, not just the final answer.
3. Where a question offers OR, attempt only one of the two choices.

| Q No. | Question | Marks |
|---|---|---|

## Level 1

8 marks

**1** MCQ, NCERT Exemplar — 1 mark

The value of $\sin^{-1}\left(\cos\frac{33\pi}{5}\right)$ is:

**(A)** $\frac{3\pi}{5}$ **(B)** $-\frac{7\pi}{5}$ **(C)** $\frac{\pi}{10}$ **(D)** $-\frac{\pi}{10}$

**2** MCQ, Board-pattern — 1 mark

The value of $\tan^{-1}\left(\cot\frac{7\pi}{5}\right)$ is:

**(A)** $\frac{7\pi}{5}$ **(B)** $\frac{2\pi}{5}$ **(C)** $-\frac{9\pi}{10}$ **(D)** $\frac{\pi}{10}$

**3** MCQ, NCERT Exemplar — 1 mark

The domain of the function defined by $f(x)=\sin^{-1}\sqrt{x-1}$ is:

**(A)** $[1,2]$ **(B)** $[-1,1]$

**(C)** $[0,1]$ **(D)** none of these

**4** MCQ, Board-pattern — 1 mark

If $x\le -1$, then the principal value $\sec^{-1}x$ lies in the interval:

**(A)** $\left[-\frac{\pi}{2},0\right)$ **(B)** $\left(\frac{\pi}{2},\pi\right]$ **(C)** $\left[0,\frac{\pi}{2}\right)$ **(D)** $\left[\pi,\frac{3\pi}{2}\right)$

**5** MCQ, Board-pattern — 1 mark

The value of $\sec^{-1}\left(-\sqrt{2}\right)-\operatorname{cosec}^{-1}\left(-\frac{2}{\sqrt{3}}\right)$ is:

**(A)** $\frac{5\pi}{12}$ **(B)** $\frac{13\pi}{12}$ **(C)** $\frac{\pi}{12}$ **(D)** $-\frac{7\pi}{12}$

| Q No. | Question | Marks |
|---|---|---|
| 6 | Assertion-Reason, CBSE SQP 2022-23<br>**Assertion (A):** The domain of the function $\sec^{-1} 2x$ is $\left(-\infty, -\frac{1}{2}\right] \cup \left[\frac{1}{2}, \infty\right)$. **Reason (R):** $\sec^{-1}(-2) = -\frac{\pi}{4}$.<br>**(A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).<br>**(B)** Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A).<br>**(C)** Assertion (A) is true, but Reason (R) is false.<br>**(D)** Assertion (A) is false, but Reason (R) is true. | 1 mark |
| 7 | Short answer, PYQ-style<br>Find the value of $\sin^{-1}\left(\sin\frac{13\pi}{7}\right) + \cos^{-1}\left(\cos\frac{13\pi}{7}\right)$. | 2 marks |

## Level 2

16 marks

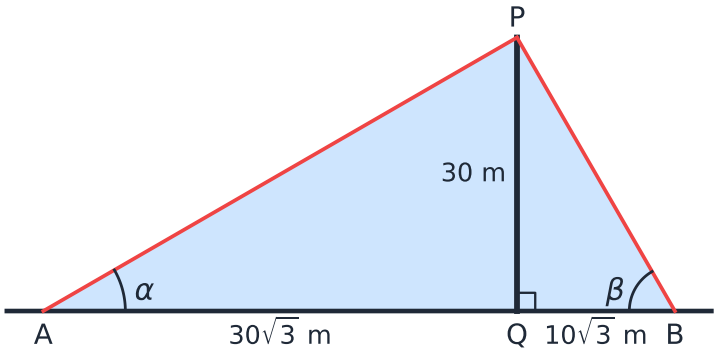
| | | |
|---|---|---|
| 8 | Short answer, Board-pattern<br>Evaluate: $\tan^{-1}\left(2\cos\frac{2\pi}{3}\right) + \sec^{-1}\left(\text{cosec}\left(-\frac{\pi}{6}\right)\right) + \cot^{-1}\left(\tan\left(-\frac{\pi}{3}\right)\right)$. | 2 marks |
| 9 | Short answer, Board-pattern<br>Find the domain of $f(x) = \sin^{-1}\left(x^2 - 2x\right)$. | 2 marks |
| 10 | Short answer, Board-pattern<br>Find the domain of $f(x) = \sin^{-1}(2x-1) + \text{cosec}^{-1}(3x)$. | 2 marks |
| 11 | Short answer, Board-pattern<br>Find the value of $\cos^{-1}\left(\cos\frac{5\pi}{4}\right) + \cot^{-1}\left(\cot\left(-\frac{\pi}{4}\right)\right)$. | 2 marks |
| 12 | Short answer, PYQ-style<br>Evaluate: $\tan\left(\frac{1}{2}\cos^{-1}\frac{\sqrt{5}}{3}\right)$. | 2 marks |
| 13 | Short answer, PYQ-style<br>If $\sin\left(\sin^{-1}\frac{1}{5} + \cos^{-1}x\right) = 1$, find the value of $x$. | 3 marks |
| 14 | Short answer, PYQ-style<br>Evaluate: $\cos\left(\sin^{-1}\frac{1}{4} + \sec^{-1}\frac{4}{3}\right)$. | 3 marks |

## Level 3

19 marks

| | | |
|---|---|---|
| 15 | Long answer, Board-pattern<br>On the same axes, draw the graphs of $y = \sin^{-1}x$ and $y = \cos^{-1}x$ for their principal value branches. Using the graphs, find the point where they meet, and write the values of $x$ for which $\sin^{-1}x > \cos^{-1}x$. | 3 marks |

| Q No. | Question | Marks |
|---|---|---|
| 16 | Long answer, Board-pattern<br>Find the value of $\tan^{-1}\left(\tan\frac{9\pi}{8}\right)+\sec^{-1}\left(\sec\frac{7\pi}{4}\right)+\operatorname{cosec}^{-1}\left(\operatorname{cosec}\frac{4\pi}{3}\right)$. | 3 marks |
| 17 | Long answer, Board-pattern<br>Find the domain and range of $f(x)=\cos^{-1}\left(\frac{x^2-1}{2}\right)$. | 3 marks |
| 18 | Long answer, NCERT Ex 2.2 Q7<br>Write $\tan^{-1}\left(\frac{3a^2x-x^3}{a^3-3ax^2}\right)$, $a>0$, $-\frac{a}{\sqrt{3}}<x<\frac{a}{\sqrt{3}}$, in the simplest form. | 3 marks |
| 19 | Long answer, NCERT Misc. Ex. Q9<br>Prove that $\cot^{-1}\left(\frac{\sqrt{1+\sin x}+\sqrt{1-\sin x}}{\sqrt{1+\sin x}-\sqrt{1-\sin x}}\right)=\frac{x}{2}$, $x\in\left(0,\frac{\pi}{4}\right)$. | 3 marks |
| 20 | Case study, Board-pattern<br>**Case Study:** A vertical tower PQ, $30$ m tall, stands on level ground with top P and foot Q. Two observers A and B stand on opposite sides of the tower, in line with Q, with QA $=30\sqrt{3}$ m and QB $=10\sqrt{3}$ m. The angles of elevation of P from A and from B are $\alpha=\tan^{-1}\left(\frac{\mathrm{PQ}}{\mathrm{QA}}\right)$ and $\beta=\tan^{-1}\left(\frac{\mathrm{PQ}}{\mathrm{QB}}\right)$. | 4 marks |




**(i)** Find $\alpha$. — 1 mark

**(ii)** Find $\beta$, and hence the angle APB at the top of the tower. — 1 mark

**(iii)** Find the values of $\sin^{-1}[\sin(\alpha+2\beta)]$ and $\cos^{-1}[\cos(\alpha-2\beta)]$. — 2 marks

# Answer key

Chapter 2: Inverse Trigonometric Functions, Practice Set B. Full step-wise solutions are in the solutions sheet.

## Level 1

| | |
|---|---|
| 1 | (D) $-\frac{\pi}{10}$ |
| 2 | (D) $\frac{\pi}{10}$ |
| 3 | (A) $[1, 2]$ |
| 4 | (B) $\left(\frac{\pi}{2}, \pi\right]$ |
| 5 | (B) $\frac{13\pi}{12}$ |
| 6 | (C) Assertion (A) is true, but Reason (R) is false. |
| 7 | $0$ |

## Level 2

| | |
|---|---|
| 8 | $\frac{5\pi}{4}$ |
| 9 | Domain $= \left[1-\sqrt{2},\ 1+\sqrt{2}\right]$ |
| 10 | Domain $= \left[\frac{1}{3}, 1\right]$ |
| 11 | $\frac{3\pi}{2}$ |
| 12 | $\frac{3-\sqrt{5}}{2}$ |
| 13 | $x = \frac{1}{5}$ |
| 14 | $\frac{3\sqrt{15}-\sqrt{7}}{16}$ |

## Level 3

| | |
|---|---|
| 15 | The graphs meet at $\left(\frac{1}{\sqrt{2}}, \frac{\pi}{4}\right)$; $\sin^{-1} x > \cos^{-1} x$ for $\frac{1}{\sqrt{2}} < x \leq 1$. |
| 16 | $\frac{\pi}{24}$ |
| 17 | Domain $= \left[-\sqrt{3}, \sqrt{3}\right]$; range $= \left[0, \frac{2\pi}{3}\right]$ |
| 18 | $3\tan^{-1}\frac{x}{a}$ |
| 19 | LHS $= \cot^{-1}\left(\cot\frac{x}{2}\right) = \frac{x}{2} =$ RHS. Hence proved. |
| 20 | (i) $\alpha = \frac{\pi}{6}$ (ii) $\beta = \frac{\pi}{3}$; $\angle \mathrm{APB} = \frac{\pi}{2}$ (iii) $\frac{\pi}{6}$ and $\frac{\pi}{2}$ |